Curvature

The KL-divergence measures the gap between two distributions. Ask what that gap looks like when the two are only a hair apart, and the answer is a *bowl*: flat at the bottom, curving upward. The curvature of that bowl is the **Fisher information**, and because it is a symmetric, positive-definite quantity that varies smoothly from place to place, it is exactly a **Riemannian metric** — a ruler on the space of distributions. This note derives the bowl, identifies its curvature as Fisher, works it out for a coin, and shows why the curvature *is* a metric.

This is a companion to *Surprise*, which built entropy, cross-entropy, and the KL-divergence from scratch and showed that a model's loss is the KL plus a constant. Everything here takes the KL as known and asks one new question: what does it look like up close?

The gap, when the two distributions are a hair apart

Fix a family of distributions controlled by a knob θ. Turning the knob sweeps out different distributions: for a coin, θ is the bias p; for a language model, the logits. Write q_θ for the distribution at setting θ, and look at the gap between the distribution at θ and the one a tiny step δ further on:

$$D_{\mathrm{KL}}\big(q_\theta \,\|\, q_{\theta+\delta}\big) \qquad \text{as a function of the small step } \delta.$$

Two facts you already know fix the shape of this function completely:

- it is **zero when** $\delta = 0$ — a distribution measured against itself has no gap;
- it is **never negative** — the KL is a gap above a floor.

A function that sits at zero and cannot go below it is sitting at the *bottom of a bowl*.

The bowl and its curvature

Here is the one fact from calculus that does all the work. *Near the bottom of a smooth bowl, every function looks like a parabola.* Two things describe it there: the bottom is **flat** (the slope is zero, which is what "bottom" means), and the sides curve upward at some rate — the **curvature**, measured by the second derivative. The flatness we already have for free; the only content left in the gap-for-a-tiny-step is *how sharply it curves*.

That curvature is the Fisher information. The whole of "infinitesimal KL = Fisher" is this single sentence:

the curvature of the KL bowl at its bottom is the Fisher information.

The rest of this section is just checking that the bowl really is a parabola and reading off its curvature.

The computation, with one knob

Write $\ell_\theta(x) = \log q_\theta(x)$ for the log-probability, and let a prime denote $\partial/\partial\theta$. Straight from the definition of KL,

$$D_{\mathrm{KL}}(q_\theta \| q_{\theta+\delta}) = \sum_x q_\theta(x)\, \log \frac{q_\theta(x)}{q_{\theta+\delta}(x)} = -\,\mathbb{E}_\theta\big[\ell_{\theta+\delta} - \ell_\theta\big].$$

Taylor-expand the bracket in the small step δ, using $\ell_{\theta+\delta} = \ell_\theta + \delta\,\ell'_\theta + \tfrac{1}{2}\delta^2\,\ell''_\theta + \cdots$:

$$D_{\mathrm{KL}}(q_\theta \| q_{\theta+\delta}) = -\,\delta\ \underbrace{\mathbb{E}_\theta[\ell'_\theta]}_{\text{first order}}\ -\tfrac{1}{2}\delta^2\ \underbrace{\mathbb{E}_\theta[\ell''_\theta]}_{\text{second order}}\ +\ \cdots$$

The first-order term vanishes, and for the same "the probabilities always sum to one" reason that runs through the whole subject:

$$\mathbb{E}_\theta[\ell'_\theta] = \sum_x q_\theta\,\frac{q'_\theta}{q_\theta} = \sum_x q'_\theta = \frac{d}{d\theta}\sum_x q_\theta = \frac{d}{d\theta}(1) = 0.$$

This is the flat bottom of the bowl, confirmed. What survives is the curvature:

$$D_{\mathrm{KL}}(q_\theta \| q_{\theta+\delta}) \;=\; \tfrac{1}{2}\,g(\theta)\,\delta^2 + O(\delta^3), \qquad g(\theta) \;=\; -\,\mathbb{E}_\theta[\ell''_\theta] \;=\; \mathbb{E}_\theta\big[(\ell'_\theta)^2\big].$$

This $g(\theta)$ is the **Fisher information**. The two expressions for it are equal: differentiate the identity $\sum_x q_\theta\,\ell'_\theta = 0$ once more with respect to θ, use $q'_\theta = q_\theta\,\ell'_\theta$, and $\mathbb{E}[(\ell')^2] = -\mathbb{E}[\ell'']$ drops out.

> **Insight** The second form, $g = \mathbb{E}[(\ell')^2]$, says what Fisher *means*. The quantity $\ell' = \partial \log q/\partial\theta$ is the **score**: how sharply the probabilities react when you wiggle the knob. Since the score averages to zero, g is its *variance* — the typical size of that reaction. Large g: the distribution lurches when you nudge θ, so nearby settings are easy to tell apart and the bowl is steep. Small g: nudging θ barely changes anything, settings are nearly indistinguishable, and the bowl is shallow. That is why it is called *information* — it is how much a single observation tells you about θ.

The coin, concretely

Take the biased coin of *Surprise*, with the knob θ being the bias p. The two outcomes have log-probabilities $\log p$ (heads) and $\log(1-p)$ (tails), so the score is $1/p$ on heads and $-1/(1-p)$ on tails, and the Fisher information is the average of its square:

$$g(p) = p\cdot\frac{1}{p^2} + (1-p)\cdot\frac{1}{(1-p)^2} = \frac{1}{p} + \frac{1}{1-p} = \frac{1}{p(1-p)}.$$

Read it off. At a fair coin ($p = \tfrac{1}{2}$) the curvature is $g = 4$, the *smallest* it ever gets: the bowl is shallowest there, and two nearly-fair coins are the hardest of all to tell apart. As p runs toward 0 or 1 the curvature $g \to \infty$: the bowl walls go vertical, and two coins near the edge — 0.99 versus 0.999 heads — are enormously far apart in KL, because the rare outcome is so informative on the occasions it finally appears.

Insight This is why the space of distributions is *curved* rather than flat. The same step δ in p counts as a tiny distance near $\frac{1}{2}$ and a huge one near the edges — the ruler itself stretches as you move around. A ruler whose length changes from place to place is exactly what a curved space is. Fisher is the thing telling you how long each little step "really" is, locally.

Many knobs: the curvature becomes a matrix

With several knobs $\theta = (\theta_1, \ldots, \theta_m)$ the bowl is multidimensional, and the single curvature number becomes a **matrix**. The same Taylor expansion gives

$$D_{\mathrm{KL}}(q_\theta \| q_{\theta+\delta}) \;=\; \tfrac{1}{2}\,\delta^\top g(\theta)\,\delta + O(\|\delta\|^3), \qquad g_{ij}(\theta) = \mathbb{E}_\theta\big[\partial_i \ell\, \partial_j \ell\big] = -\,\mathbb{E}_\theta\big[\partial_i \partial_j \ell\big],$$

the **Fisher information matrix**. The quadratic form $\delta^\top g\,\delta$ is the squared gap for a displacement δ in any direction through parameter space.

From curvature to a Riemannian metric

Now the last step: why this curvature is not merely *like* a ruler but actually *is* one.

Stripped of jargon, a **Riemannian metric** is a rule for measuring the length of a tiny displacement, with two features: it is allowed to *change from point to point*, and at each point it is a genuine squared-length rule — a symmetric, positive-definite quadratic form (an inner product). Compare that against what we have. The gap for a small step is $\delta^\top g(\theta)\,\delta$, a squared length for the displacement δ, and $g(\theta)$ is:

- **symmetric** — $g_{ij} = g_{ji}$ on sight;
- **positive-definite** — it is a covariance matrix, $\mathbb{E}[\,\mathrm{score} \cdot \mathrm{score}^\top]$, hence never negative, and strictly positive in every direction that genuinely changes the distribution;
- **smoothly varying** — the coin's $1/(p(1-p))$ changes as p moves, and so does g in general.

Those three properties *are* the definition of a Riemannian metric. So the Fisher information is a metric tensor on the space of distributions. That space becomes a genuine curved surface, and the length of any path across it is obtained by adding up $\sqrt{\delta^\top g\,\delta}$ along the way; the shortest such path is the **Fisher–Rao distance**, the natural, coordinate-free distance between two distributions.

Insight This resolves the nagging business of calling the KL a "distance" when it is not. *Globally*, KL is lopsided — $D_{\mathrm{KL}}(P\|Q) \neq D_{\mathrm{KL}}(Q\|P)$, no triangle inequality, not a distance. *Infinitesimally*, that asymmetry is a *third*-order effect: it lives entirely in the $O(\delta^3)$ terms we discarded, so to second order the bowl is perfectly symmetric and hands back a true, symmetric metric. KL in the large is a one-way gap; its shadow in the small is a clean two-way ruler, and the Fisher metric is exactly that shadow.

Insight One fact worth knowing, even without its proof. On the space of probability distributions, the Fisher metric is the *unique* Riemannian metric — up to an overall scale — that is invariant under the operations that lose no information (Chentsov's theorem). You do not get to choose a metric and the geometry is not a convention: once you ask for

distances between distributions that do not depend on how you happened to coordinatise them, it is Fisher–Rao or nothing.

A metric on the meaning manifold

One caution before the victory lap. Everything so far lives on the space of *distributions*. But the "meaning manifold" of *Next-Token Prediction is Geometry* lives somewhere else: it is the space of hidden states $h \in \mathbb{R}^d$, the context compressed to a point. That space has no canonical ruler of its own. The Euclidean inner product $\langle h, e_w \rangle$ it came with is an accident of the coordinates the network happened to learn — two hidden states can be Euclidean-near yet predict entirely different next words. So we do not *find* a metric on hidden-state space; we **borrow** one.

The borrowing has a name, *pullback*, and a single idea behind it: to measure the distance between two hidden states, do not compare them directly — compare the *distributions they produce*, with the Fisher metric those distributions already carry. Every hidden state h emits an output distribution $q_h = \mathrm{softmax}(Wh)$, and a small step dh produces a small change $dq = J\,dh$ in it, where $J = \partial q / \partial h$ is the Jacobian of the map. Fisher measures that change as $dq^\top g\, dq$, so the length of dh is

$$dh^\top \big(J^\top g\, J\big)\, dh, \qquad \hat{g}(h) = J^\top g\, J.$$

That sandwich — Jacobian, Fisher, Jacobian — is the pullback. Pushed through the logits $z = Wh$ (so $\partial z / \partial h = W$, and the Fisher metric in logit coordinates is $\mathrm{diag}(p) - pp^\top$) it becomes the concrete metric on hidden-state space:

$$\hat{g}(h) \;=\; W^\top \big(\,\mathrm{diag}(p) - p\,p^\top\big) W, \qquad p = \mathrm{softmax}(Wh),$$
$$D_{\mathrm{KL}}\big(q(h)\,\|\,q(h+dh)\big) \;=\; \tfrac{1}{2}\, dh^\top \hat{g}(h)\, dh + \cdots .$$

It is the bowl of this whole note re-raised on hidden-state space: the Fisher length of a step dh is the KL gap between the next-token distributions before and after the step.

Insight Read what $\hat{g}$-distance means: two hidden states are close exactly when they induce *similar next-token distributions*. That is the distributional hypothesis — *you shall know a word by the company it keeps* — turned from slogan into a Riemannian metric. And it is *not* the Euclidean distance on h; it is Euclidean distance warped by how sharply the output reacts, the warp factor being Fisher at p. The flat inner-product picture of the first note was a chart; this is the territory.

The construction is general, not a language-model trick: any map $f : M \to N$ from a manifold into a Riemannian manifold (N, g) induces a **pullback metric** on M by $(f^* g)(u, v) = g\big(df\,u,\, df\,v\big)$ — measure the image of a vector, and call that the length of the vector. Here M is hidden-state space, N is the distributions with their Fisher metric, and $f = \mathrm{softmax} \circ W$. Equivalently — and this is "a metric on the sub-manifold" said directly — the image $\{q_h\}$ is a submanifold of the simplex and inherits Fisher by *restriction*; the pullback to h-space and the restriction to the image are the same metric, written in two charts.

Two honest caveats finish the picture.

- **It can be degenerate.** A step dh that shifts every logit by the same amount leaves the distribution unchanged — the softmax gauge freedom — so it has zero $\hat{g}$-length. The metric is

strictly positive-definite only on the directions that actually move the output; the invisible directions are genuinely invisible to this ruler.

- **Flat ambient, curved slice.** The full simplex is dually flat, but the image the network sweeps out is in general a *curved* submanifold of it; its embedding curvature measures how far the model's own family is from being an exponential family in its own right.

The chain, in one line

KL measures a gap → look at the gap for a tiny step → it is a bowl with a flat bottom → the bowl's curvature is the Fisher information → that curvature, symmetric and positive-definite and smoothly varying, is a Riemannian metric → the space of distributions is a curved surface with Fisher as its ruler → pull that ruler back through the softmax and the model's hidden states inherit it, a metric under which "near" means "predicts alike."